

Open, Slow, Green and Inclusive Science: Adoption and Barriers to New Research Practices [TENTATIVE TITLE]

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Materials:

 github.com/DominiqueMakowski/OpenScienceArchetypes
 osf.io/2es9b

Open science



Dominique Makowski¹ , **Julia Ayache²**, **Axel Garnier³**, **Louise Kirsch²**, **Marco Sperduti^{4,5}** & **Klara Kovarski^{6,7}**

¹School of Psychology, University of Sussex, Brighton, UK.

²INCC, Université Paris Cité, France.

³LINC – Laboratoire de recherches Intégratives en Neurosciences et psychologie Cognitive, Inserm UMR1322, Besançon, France.

⁴University of Rome Tor Vergata, Italy.

⁵LMC2 (Laboratoire Mémoire, Cerveau et Cognition), Université Paris Cité, France.

⁶LaPsyDé, CNRS & Université Paris Cité, France.

⁷Faculté de Lettres, INSPE, Sorbonne Université, France.

Open science is one of several movements calling for a reform of research practices, alongside slow, sustainable, and ethical and inclusive science. Whether researchers endorse them as one project is unknown. We surveyed 672 researchers working in Europe, mostly in France, at all career stages, on their knowledge, attitudes and adoption of the practices these movements promote. Attitudes and practices formed five facets in two nearly independent sides: an epistemic side, which separated the sharing of research outputs from the reform of procedures such as preregistration, and a societal side, which merged environmental and ethical concerns. Slow science bridged the two sides, although its link with the epistemic side mostly reflected familiarity with the movement. Career stage shaped which reforms researchers carried more than how much they endorsed reform: PhD students shared fewer outputs, preregistration was more common before than after obtaining a permanent position, and every stage asked for structural levers, such as career recognition, rather than information. Women scored higher than men on the societal facets, increasingly with age. Five profiles summarised how researchers combine the two sides. Reform agendas need to link the reliability and the responsibility of research rather than assume this link.

Introduction

Over the past two decades, the reliability of published findings has become a central concern in several empirical disciplines, especially psychology, the social sciences and biomedicine (Ioannidis, 2005; Open Science Collaboration, 2015). Large-scale replication projects and surveys on questionable research practices (John et al., 2012) showed that part of the published evidence is less robust than assumed. Beyond methodological issues, this “replication crisis” opened a broader reflection on how research

The study and its analysis plan were preregistered on OSF (<https://osf.io/2es9b>). The data and analysis code are available at <https://github.com/DominiqueMakowski/OpenScienceArchetypes> and will be archived on Zenodo. The authors have no competing interests to declare. This study was not supported by dedicated funding. We thank the colleagues all over Europe who accepted to share this survey within their laboratories and communities.

✉ e-mail: D.Makowski@sussex.ac.uk

is organised: publication systems, evaluation criteria and career incentives (Ioannidis et al., 2015; Lilienfeld, 2017).

The most visible response has been open science, an umbrella term for practices that make the research process more transparent and accessible to anyone, in academia and beyond, such as preregistration, the sharing of data, materials and code, and open access publication (Fecher & Friesike, 2014; Pontika et al., 2015). Open science is supported at the institutional level in many countries. The San Francisco Declaration on Research Assessment calls for evaluating research on its content rather than on journal metrics (San Francisco Declaration on Research Assessment, 2012), and the European Commission has made open access a condition of funding (Guedj & Ramjoué, 2015), so that open science practices are increasingly valued in research careers (Cabello Valdes et al., 2017). In France, the national plans for open science and the open archive HAL encourage, and sometimes require, the deposit of scientific publications (Ministère de l'Enseignement supérieur, de la Recherche et de l'Innovation, 2021). Despite this support, adoption remains heterogeneous, and open science can also be perceived as an additional, time-consuming task. Most European researchers surveyed in 2017 were not familiar with the concept of open science (Cabello Valdes et al., 2017). Awareness has increased since then (Christensen et al., 2019), but practices are adopted at very different rates, open access for instance much more than preregistration (Norris et al., 2022), a practice that has to be performed before the data are collected or analysed.

Open science is not the only reform-oriented framework, and several other concerns have emerged in the scientific community. The slow science movement criticises the pressure to publish fast and argues that a slower pace favours deeper and more reliable work, at the cost of publishing less (Frith, 2020; Stengers, 2018); Frith (2020) even proposed to limit the number of publications per researcher per year. In a similar vein, sustainable (or green) science addresses the environmental and societal costs of research, such as the carbon footprint of travel and computing (Mariette et al., 2022). Finally, ethical and inclusive science focuses on equity, diversity and representation in research teams and samples, for instance the participation of under-represented groups in the scientific process, such as persons with disabilities, or women in the disciplines where men still predominate (Fox et al., 2021). These frameworks are mostly discussed separately, although they share concerns about research quality and responsibility.

Their compatibility is not trivial. Some open science practices fit well with slow science: preregistration and replication require time and favour quality over quantity, as they require anticipating the hypotheses and analyses. Other practices might be questioned. Preprints increase the number of shared articles, and open science takes no explicit position on publication volume, with consequences for the amount of information available and for the environmental footprint of research. Similarly, article pro-

cessing charges for open access may exclude researchers with fewer resources, in tension with inclusivity goals (Fox et al., 2021), and international collaboration and conference attendance conflict with the carbon reduction called for by sustainable science. Whether researchers perceive these frameworks as one coherent reform or as competing demands has, to our knowledge, not been investigated.

In this article, we refer to open, slow, sustainable, and ethical and inclusive science as reform *movements*, i.e., the frameworks under which reform-oriented practices are promoted and discussed. We refer to the dimensions along which researchers' attitudes and practices actually vary as the *facets* of reform-oriented science. Facets and movements need not coincide: a movement may split into several facets if its practices are adopted independently of one another, and several movements may merge into a single facet if the same researchers endorse them together. Whether reform-oriented science is one project with several facets, or a set of separate movements, is thus an empirical question.

Career stage and employment conditions are likely central in these dynamics. Early-career researchers are often presented as drivers of open science (Farnham et al., 2017; Nicholas et al., 2019) and receive more training in it during their PhD or early career. Yet they are less involved in decision making at the institutional level, from the laboratory to the university, in grant applications and in publication strategy (Allen & Mehler, 2019), and are more exposed to precarious contracts and to the "publish or perish" pressure (Enright & Facer, 2017; Herschberg et al., 2018). Surveys consistently report that early-career researchers value open science but apply it only partly, and that lack of time and lack of institutional support are the main barriers (Pownall et al., 2023; Toribio-Flórez et al., 2021; Zečević et al., 2020). These studies, however, share several limits: they focus mainly or only on open science, they sample early-career researchers without comparison to later career stages, they are often restricted to one institution or country, and they do not distinguish career stage from precariousness. It therefore remains unclear which barriers are specific to career stage, which are specific to precarious employment, and how attitudes towards open science relate to the other movements.

The present study

The present study addresses these questions with an online survey of researchers at all career stages, from PhD students to emeritus researchers, involved in experimental research in the social sciences, life sciences and physical sciences. The survey targeted researchers working in Europe, with a particular focus on France, where most of the authors are based. We measured knowledge of, attitudes towards and self-reported adoption of the practices promoted by four reform movements, open, slow, sustainable and ethical and inclusive science, as well as the perceived barriers to open science. We tested four preregistered hypotheses: (i) early-career researchers report more knowledge of and training in open science than senior researchers; (ii) this

difference does not extend to adoption, because adoption also depends on autonomy; (iii) lack of time and lack of institutional support are the most reported barriers to open science, replicating previous findings; and (iv) barriers related to time and evaluation pressure are more reported by non-permanent researchers. We expected attitudes towards the four movements to be positively correlated, and complemented this test with exploratory analyses of the relationships between practices, to identify the facets that structure them and to reveal potential contradictions, and of how these facets relate to career stage, employment status and perceived precariousness. In addition, we explored the role of meta-scientific training and the differences between disciplines, and compared the respondents working in France with those working in other European countries, the second objective of the preregistration.

Methods

Participants

A total of 682 participants completed the survey. As the survey targeted researchers working in Europe, the 10 participants (1.5%) who reported a workplace in another continent (Australia, Brazil, Canada, China, Puerto Rico, United States) were excluded, which left 672 participants. Participants had a mean age of 38.1 years ($SD = 11.9$; 2 implausible ages, below 18, were set to missing) and included 384 women, 278 men, 9 non-binary participants and 1 participant who reported another gender. Due to the small sample of non binary participants, analyses on gender were carried on only male and female participants. Participants worked in 17 European countries, mostly in France (72.9%), followed by the United Kingdom (8.5%), Italy (5.1%) and Germany (3.1%). They worked across the three research domains (Social Sciences & Humanities: 49.3%; Physical Sciences & Engineering: 18.2%; Life Sciences: 32.6%). Regarding career stage, 42.9% held a permanent position, 25.7% were PhD students (funded or self-funded), 12.5% were postdoctoral researchers, 9.4% were researchers or lecturers on fixed-term contracts, 3.9% were engineers on fixed-term contracts, 3.0% were master students or research assistants, and 2.7% reported another position. Completion times and response patterns gave no sign of careless responding: 0.6% of participants completed the survey in less than five minutes, and none gave the same rating to all the sliders of the four movements. No participant was excluded on these grounds.

Survey

Data were collected using an online survey (LimeSurvey from Sorbonne Université) targeting researchers currently involved in experimental research, either in permanent or non-permanent positions. The survey took a median of 11.6 minutes to complete and was distributed anonymously; data were collected with the intention of open sharing with the scientific community. Ethical approval was obtained from Université Paris Cité (IRB: 00012024-29). The study and analyses plan were preregistered on OSF platform (<https://osf.io/2es9b>), and the

data and analysis code are available at <https://github.com/DominiqueMakowski/OpenScienceArchetypes> and will be archived on Zenodo.

The questionnaire included 39 questions, some of them grids of several items, of which each participant answered 35 to 37 depending on their position and previous answers (e.g., the questions on career prospects differed between permanent and non-permanent researchers). They were organized into the following sections: i) *Demographics and career information*: age, gender, country of workplace, academic affiliation, discipline (i.e., Social Sciences & Humanities, Physical Sciences & Engineering, Life Sciences) and subdiscipline (not mandatory), highest diploma obtained, PhD starting year, current contract type (e.g., PhD student, postdoc, permanent position), additional professional activities (e.g., clinical, committees), and the approximate distribution of working time across research, teaching, administration, popularization, and other activities. ii) *Career perceptions*: intention to pursue an academic career (for non-permanent researchers only), estimated probability of obtaining a permanent position within an acceptable timeframe, publication number, job title (for permanent researchers), estimated likelihood of career progression, and satisfaction with the number and quality of one's publications. iii) *Criteria for research quality*: participants selected up to three criteria they considered most important for assessing the quality of scientific work (e.g., innovation, methodological rigor, transparency, replication, environmental impact, inclusivity).

Then, three randomized blocks were presented, on open, slow and sustainable science, followed by the items on ethical and inclusive science [TO CONFIRM: the revised text mentions three randomized blocks and lists ethical/inclusive science as a fourth part; was it randomized with the others?].

Open science. Participants rated their familiarity with the open science movement and its perceived importance, and indicated whether they had taken part in a related training. They then reported their knowledge and use of specific practices (preregistration, registered reports, open materials, open data, open peer review, open access, replication studies, participatory research), choosing between "I am not familiar with it", "I haven't used it yet, and I don't plan to do it in the future", "I haven't used it yet, but I plan to do it in the future" and "I know it and I used it". Finally, they selected up to five factors that would facilitate their further adoption of open research practices.

Slow science. Participants rated their familiarity with the slow science movement and its perceived importance, indicated whether they had taken part in a related training, and selected up to three statements that best define slow science for them (e.g., quality over quantity, longer timescales, fewer publications). They then rated the feasibility (at the individual level, at the institutional level, or not feasible) of specific measures (e.g., limiting the

number of submitted articles, replicating before publishing, valuing teamwork over individual output).

Sustainable science. Participants rated the importance of environmental sustainability in how they conduct their research and in how they choose their topics, the extent to which they had changed their research practices (e.g., experimental design, material) and their communication practices (e.g., professional travel) for environmental reasons, the link they perceive between scientific production and environmental sustainability, and their willingness to change their practices for environmental reasons.

Ethical and inclusive science. Participants rated the importance they attribute to the diversity of the research team (e.g., in terms of neurodiversity, disability, ethnicity or gender), and to being careful about the direct consequences (e.g., benefits or harms) of their research for participants and society.

As a conclusion, general wellbeing and alignment questions were presented: overall agreement with statements regarding job fulfillment, alignment between research practices and personal principles, sufficiency of time for research, work/life balance satisfaction, and career-related worry.

Most attitudinal items were rated on continuous sliders ranging from 0 (“not at all”) to 100 (“extremely”), while categorical items used single- or multiple-choice formats. Depending on hypotheses and research questions, some questions were presented as one choice questions, while others as multi-answers (but with constraints, e.g., selecting up to 3 choices).

Procedure

The survey was shared through multiple channels to reach a broad and diverse sample of researchers. First, it was distributed within the universities and research institutes affiliated with the authors. Second, it was shared through several academic mailing lists and domain groups. Third, European researchers working in the fields under investigation (open science, environmental research, slow science...) were directly contacted and asked to relay the survey within their own professional networks.

The initial target sample size was set at 3,000 participants. However, after one year of active dissemination, the research team encountered substantial recruitment difficulties and was unable to reach this target. The team therefore decided to proceed with the planned analyses after verifying that the collected sample met three minimal criteria: a sufficient number of participants across the three research domains (Social Sciences & Humanities, Physical Sciences & Engineering, and Life Sciences), a satisfactory balance between permanent and non-permanent researchers, and a satisfactory balance between men and women, allowing to test most hypotheses.

Data analysis

The analysis followed three axes. First, we described the landscape of research practices (percentage of participants selecting each option; mean, M , and standard deviation, SD , of the 0–100 sliders), and tested hypotheses (i) to (iv) by modelling each indicator of knowledge, training, adoption and barrier on career stage, in three levels: PhD students (with master students and research assistants), non-permanent positions (postdocs, and researchers, lecturers and engineers on fixed-term contracts) and permanent positions (654 participants). In what follows, *early-career researchers* refers to the first two levels and *senior researchers* to the third.

Second, we examined the structure of the 19 rating items of the four movements, all scaled to 0–1: the importance of open science and the knowledge and use of eight open science practices, the familiarity with and importance of slow science, the six items on environmental sustainability and the two items on ethical and inclusive science. The knowledge and use of each practice was scored as a level of engagement (“I am not familiar with it”: 0; “I haven’t used it yet, and I don’t plan to do it in the future”: 1/3; “I haven’t used it yet, but I plan to do it in the future”: 2/3; “I know it and I used it”: 1), unfamiliarity being placed below deliberate non-use because participants who did not know a practice were less familiar with, and less often trained in, open science than those who knew it but did not plan to use it. The number of factors was chosen by the agreement between 19 methods (Lüdtke et al., 2020), the structure was explored with an exploratory factor analysis [EFA; oblimin rotation; Revelle (2026)] and tested with confirmatory factor analyses (CFA) (Rosseel, 2012), the retained one providing the facet scores. The structure of the nine criteria of scientific quality was examined with a principal component analysis (PCA). The facet structure held under alternative scorings of the practices and when the practice items were treated as ordinal, and its measurement invariance across gender, career stage and discipline was tested (Chen, 2007); these analyses are reported in the supplementary materials.

Third, the facet scores were modelled with Bayesian regressions (Bürkner, 2017): on gender and age (second-degree polynomial, in interaction; 654 women and men aged 65 or less), on career stage (and, as sensitivity analyses, on career stage in interaction with career worry or with the time available for research), and, to examine what lies behind these effects, on career stage with training in open and in slow science as covariates, on discipline, on gender and age with discipline as a covariate, on the perceived probability of obtaining a permanent position (among the 248 non-permanent researchers), and on region (France vs. other European countries, with and without career stage and discipline as covariates). The seven ratings of career perceptions and well-being were modelled on the five facets and career stage. Researcher profiles were identified by k-means clustering of the standardised facet scores (Hartigan & Wong, 1979), retaining the partition with the lowest within-cluster sum of squares over 200

random starts, the number of clusters being examined with the agreement between 24 methods (Charrad et al., 2014) and the stability of the solution by bootstrap, as a whole (adjusted Rand index, Hubert & Arabie, 1985) and profile by profile (Jaccard similarity, Hennig, 2007). To tell groups from a continuum, these diagnostics and the average silhouette width (Rousseeuw, 1987) were compared with those of ten datasets simulated without groups, from a Gaussian copula with the marginal distributions and rank correlations of the facet scores. Profile membership was modelled with a Bayesian categorical regression on gender and age, and with multinomial models adding discipline and region. We report posterior medians and 95% credible intervals (CI) of marginal means, contrasts and average slopes (Makowski et al., 2025), in units of facet score unless stated otherwise. Priors, diagnostics and all intermediate results are given in the analysis notebook (see the supplementary materials).

Results

The Landscape of Research Practices

What makes good science?

Asked to select the three most important criteria to assess the quality of scientific work (Figure 1 A), participants overwhelmingly chose a rigorous methodology (83.0%), well ahead of transparency and open science (46.0%), appropriate statistical analyses (43.2%), originality and innovation (41.5%), and significance and impact (40.0%). Publication in high impact factor journals was chosen by only 7.1% of participants, as rarely as inclusivity and diversity (6.8%) or environmental impact (6.7%). Participants thus located the quality of science in the soundness of its process rather than in the prestige of its outlets, and rarely in its social and environmental aspects, although these were highly rated elsewhere in the survey.

Four movements, unequally known

The four movements differed first in how well they were known (Figure 1 B). Open science was familiar to most participants ($M = 73.5$, $SD = 24.2$) and rated as important ($M = 82.1$, $SD = 22.6$), and 53.0% had attended a training or workshop on it. Slow science was much less known ($M = 34.0$, $SD = 32.7$): 32.6% of participants reported no familiarity at all with the movement, and only 5.4% had been trained in it. As the question on its importance was mandatory, nearly all the participants who did not know the movement answered 0 (89.5%, vs. 10.2% of the familiar participants). We read these answers as an inability to rate the movement, and describe its importance among the 453 familiar participants, who rated it as fairly important ($M = 64.3$, $SD = 30.5$).

Environmental sustainability was endorsed in principle more than in practice. Participants were willing to change their research practices for environmental reasons ($M = 72.7$, $SD = 25.0$), saw a link between scientific production and environmental sustainability ($M = 60.9$, $SD = 29.2$), and considered sustainability important in how they conduct research ($M = 59.3$, $SD = 27.4$), less so in how they

choose their topics ($M = 49.7$, $SD = 30.3$). The changes they reported having made were more modest, especially to research practices such as experimental design or material ($M = 36.4$, $SD = 30.5$; 22.0% reported no change at all) compared with communication practices such as professional travel ($M = 54.7$, $SD = 33.7$). Finally, ethical considerations were highly rated, in particular being careful about the consequences of one's research for participants and society ($M = 82.7$, $SD = 20.1$), and to a lesser extent the diversity of the research team ($M = 70.1$, $SD = 26.6$).

Open science: sharing outputs is mainstream, reforming procedures is not

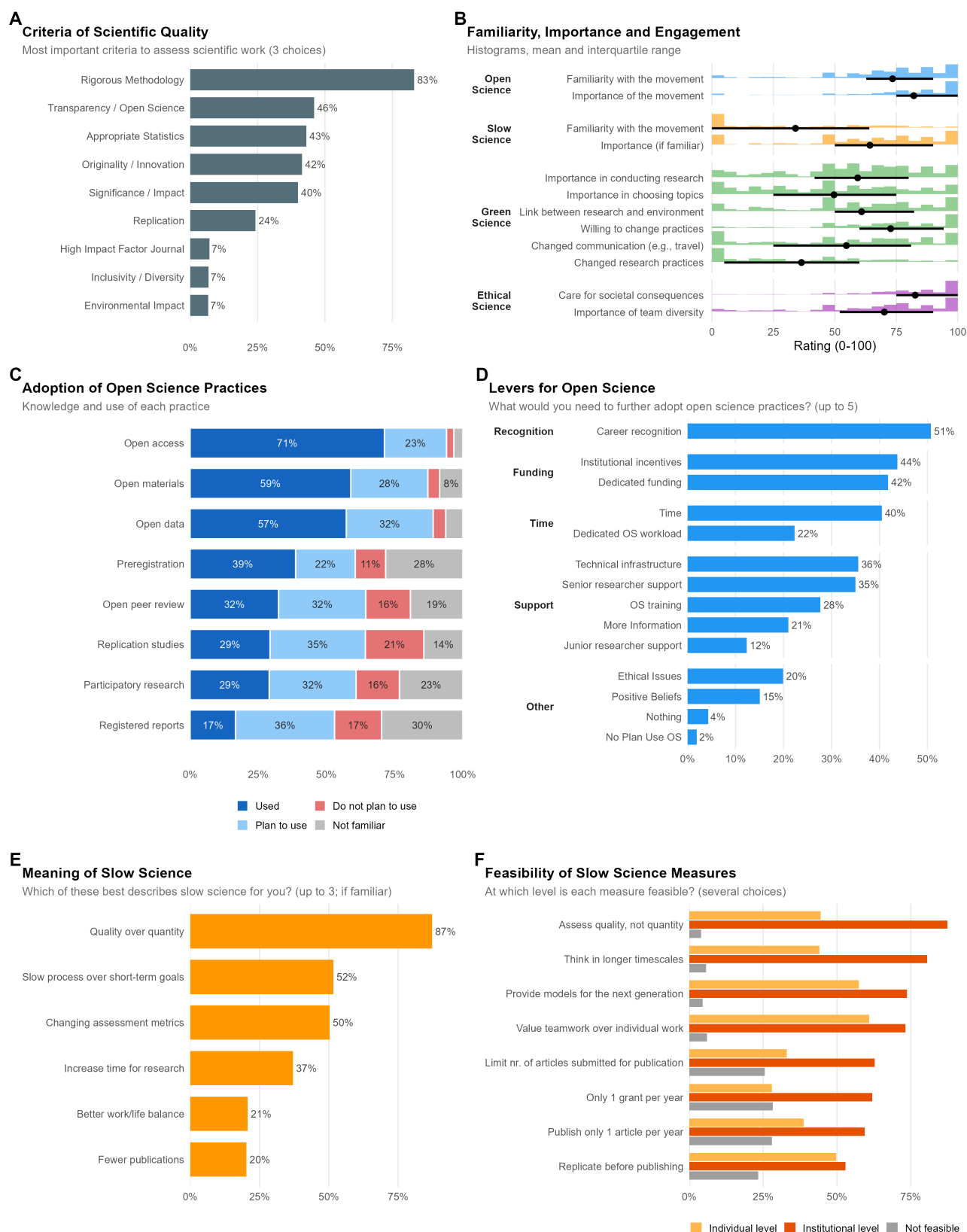
The adoption of open science practices depended on whether they concern the outputs or the process of research (Figure 1 C). Sharing outputs was widespread: 71.3% of participants had published in open access, 58.9% had shared their materials and 57.3% their data. Practices that change how research is conducted and evaluated were less common: preregistration (38.7%), open peer review (32.4%), replication studies (29.3%), participatory research (29.0%) and registered reports (16.7%). This gap reflected unfamiliarity and unrealised intentions more than rejection: registered reports and preregistration were unknown to 29.8% and 28.3% of participants respectively, 36.3% planned to use registered reports, and few participants did not plan to use a practice they knew (at most 21.4%, for replication studies).

When asked what would help them adopt open science practices further (Figure 1 D), participants mostly pointed to structural levers: career recognition in recruitment and promotion (50.7%), incentives from funders and institutions (43.8%), dedicated funding (41.8%) and time (40.5%), followed by technical infrastructure (35.6%) and the support of senior researchers (35.0%). Training (27.7%) and information (21.0%) were selected less often, and only 4.3% of participants reported that nothing would help. The barriers to open science are thus perceived as lying in the incentive structure of research rather than in a lack of knowledge or motivation, even though a sizeable minority of participants did not know procedural practices such as preregistration.

Slow science: quality over quantity, a task for institutions

Asked what best describes slow science (Figure 1 E), most of the participants familiar with the movement chose "quality over quantity" (87.2%), followed by favouring slow processes over short-term goals (51.7%) and changing assessment metrics (50.3%); fewer associated it with more time for research (37.1%), a better work/life balance (20.8%) or publishing less (20.3%).

Every slow science measure was more often judged feasible at the institutional than at the individual level (Figure 1 F), most of all assessing the quality rather than the quantity of research (institutional: 87.4%, individual: 44.5%) and thinking in longer timescales (80.5% vs. 44.0%). Measures that cap research output were the

Figure 1*Survey Responses on the Quality of Science and on Open, Slow, Green and Ethical Science*

Note. **A:** Percentage of participants selecting each criterion among the three most important to assess the quality of scientific work. **B:** Distribution of the 0–100 slider ratings (histograms), with their mean (point) and interquartile range (line); the importance of slow science is shown among the participants familiar with the movement. **C:** Knowledge and use of open science practices. **D:** Percentage of participants selecting each avenue (up to five) that would help them further adopt open science practices. **E:** Percentage of the participants familiar with slow science selecting each of its definitions (up to three). **F:** Percentage of participants judging each slow science measure feasible at the individual level, at the institutional level (both could be selected), or not feasible. Percentages are of all participants, except in E.

most contested: limiting the number of submitted articles, publishing only one article per year and applying for only one grant per year were judged not feasible by 25.6%, 28.0% and 28.3% of participants respectively, as was replicating one's results before publishing them (23.4%), against 4.5–6.0% for the other measures. Slow science is thus understood as a change in how research is valued, to be carried by institutions, rather than as a reduction of output to be achieved by individuals.

The Facets of Reform-Oriented Science

Four movements, five facets

The agreement between methods favoured five factors (supported by 5 of the 19 methods, the largest number), which together explained 45% of the variance of the items and did not map one-to-one onto the four movements (Figure 2 A). Facet names are capitalised hereafter (e.g., Open Science) to distinguish them from the movements (e.g., open science).

First, the open science movement split into two facets. *Open Science* gathered the sharing of research outputs: open data (.71), open access (.59) and open materials (.54), together with the importance attributed to the movement (.34). *Rigorous Science* gathered practices that reform how research is designed and evaluated: registered reports (.77), preregistration (.73) and, more weakly, replication studies (.46) and participatory research (.43). Open peer review loaded weakly on all factors (at most .27). Sharing one's outputs and changing one's procedures are thus distinct commitments, in line with their very different rates of adoption (Figure 1 C).

Second, the slow science movement formed a single facet, *Slow Science*, defined by the familiarity with the movement (.94) and its perceived importance (.84). As the participants who did not know the movement could not rate its importance, this facet measures *engagement* with slow science, i.e., knowing and valuing it, just as the open science facets combine knowing and using each practice. Treating the importance ratings of the unfamiliar participants as missing barely changed the scores ($r = .92$ with the retained ones), as their familiarity rating stays at 0.

Third, the environmental and ethical movements partly merged. *Green Science* gathered the importance of sustainability in how research is conducted (.77) and in the choice of topics (.71), and the changes made to research (.68) and communication practices (.50). *Ethical Science* gathered the importance of team diversity (.54) and the care for the societal consequences of research (.50), and two environmental items loaded on both facets: the perceived link between research and environmental sustainability (Ethical Science: .43; Green Science: .30) and the willingness to change one's practices for environmental reasons (Ethical Science: .41; Green Science: .43). Ethical Science thus describes a *responsible* science more broadly, extending to the consequences of research, including environmental ones, whereas Green Science is anchored in sustainability as it is put into practice.

A CFA assigning the items to the facets by movement (with open science split as above, and open peer review on Rigorous Science) fitted the data only moderately (CFI = .891, TLI = .869, RMSEA = .063). Letting the two environmental beliefs also load on Ethical Science, as in the EFA, improved the fit ($\Delta\chi^2(2) = 100.5$, $p < .001$), more than moving them to Ethical Science only ($\Delta\chi^2(2) = 28.9$, $p < .001$). This model fitted acceptably (CFI = .920, TLI = .902, RMSEA = .054 [.048, .060], SRMR = .053) and provided the facet scores. As its cross-loadings were suggested by the EFA of the same data, it describes the structure of the facets rather than confirming it independently.

Epistemic and societal sides, loosely bridged by slow science

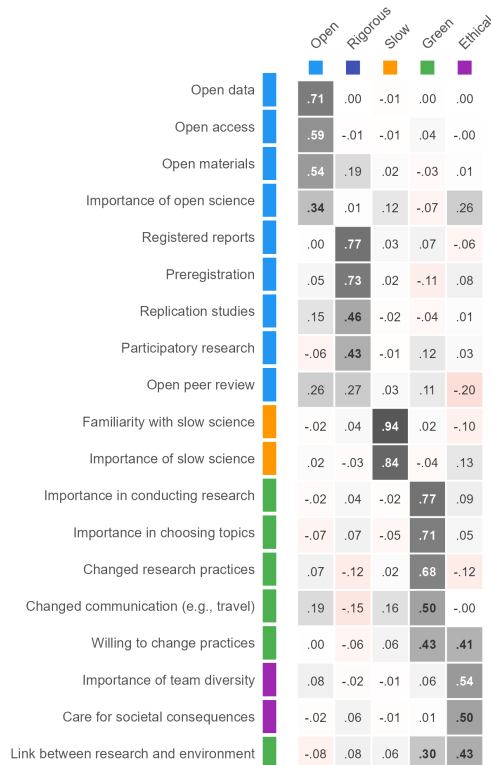
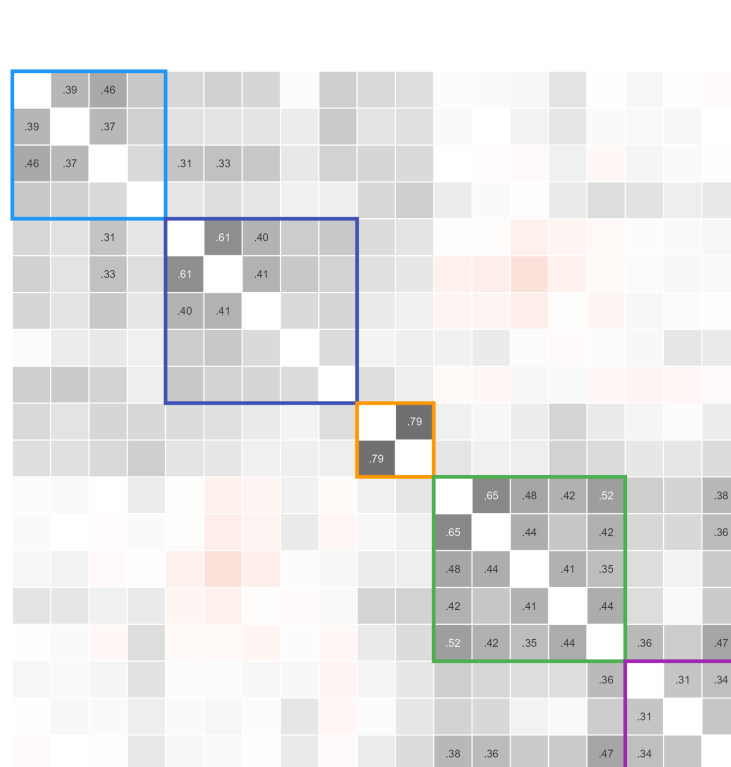
The correlations between the facets (Figure 2 C) revealed two pairs. The two open science facets were correlated ($r = .53$ [.44, .61]), forming an epistemic side concerned with how research is conducted and shared, and Green and Ethical Science were correlated to the same extent ($r = .52$ [.40, .63]), forming a societal side concerned with the consequences of research. The two sides were nearly independent: Green Science was unrelated to Open Science ($r = .05$ [-.05, .15]) and slightly negatively related to Rigorous Science ($r = -.12$ [-.22, -.03]), and Ethical Science was unrelated to both ($r = .06$ [-.05, .18] and .06 [-.05, .17]). Slow Science stood between them, related to Open ($r = .34$ [.25, .42]) and Rigorous Science ($r = .20$ [.11, .28]) as well as to Green ($r = .18$ [.09, .26]) and Ethical Science ($r = .25$ [.15, .34]). This bridge, however, rests partly on knowing the movement. Among the 453 participants at least somewhat familiar with slow science, its correlation with Rigorous Science vanished ($r = .01$) and that with Open Science halved ($r = .16$), whereas those with Ethical ($r = .31$) and Green Science ($r = .14$) held. Endorsing the reform of research practices is thus largely independent of endorsing their environmental and ethical responsibility. Slow science connects the two sides, but its link with the epistemic side mostly reflects exposure to the debates on research practices, whereas its link with the societal side reflects the endorsement of its values.

The structure of quality criteria

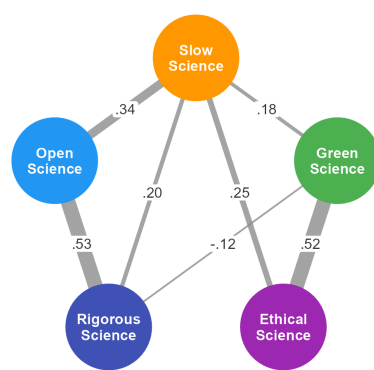
A related tension appeared in the criteria of scientific quality (Figure 2 D). As each participant selected three criteria, the criteria were mostly negatively correlated. The first principal component (20% of the variance) opposed novelty and impact (originality, significance, high impact factor journals) to rigour and transparency, and the second (14%) opposed the classical criteria of good research, rigour and innovation, to those carried by the reform movements: transparency, inclusivity, environmental impact and replication. Part of these oppositions is built into the forced choice, but they indicate that, when quality must be defined in a few words, the values of the reform movements compete with the novelty and impact that traditionally define it. The criteria chosen also converged

Figure 2*The Facets of Reform-Oriented Science***A Facets (Exploratory Factor Analysis)**

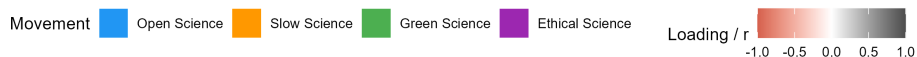
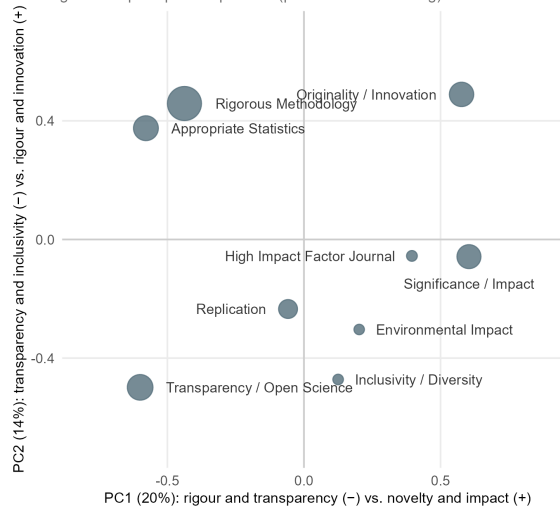
Loadings of each item, by the movement it was asked under

**B Item Correlations**Items in the same order as in A; $|r| \geq .30$ labelled; facets framed**C****Correlations between Facets**

Latent correlations (CFA) whose 95% CI excludes 0

**D****Structure of Quality Criteria**

Loadings on the principal components (point size: % selecting)



Note. **A:** Loadings of the 19 rating items on the five factors of the exploratory factor analysis (oblimin rotation), with the movement under which each item was asked (left strip). Items are ordered by the facet on which they load most, and loadings of at least .30 are in bold. **B:** Correlations between the same items, in the same order, with the items loading most on each facet framed; correlations of at least .30 in absolute value are labelled. **C:** Correlations between the facets estimated by the confirmatory factor analysis, shown when their 95% confidence interval excludes 0. **D:** Loadings of the nine criteria of scientific quality on the first two components of a principal component analysis; point size is proportional to the percentage of participants selecting the criterion.

Figure 3
Open Science Practices, Levers and Facets by Career Stage



Note. **A:** Probability of having used each open science practice, by career stage, from Bayesian logistic regressions. Practices are grouped by the facet on which they load (Figure 2 A). **B:** Probability of selecting each lever that would help adopting open science practices (up to five), by career stage, grouped as in Figure 1 D. **C:** Facet scores by career stage (Bayesian linear regressions, marginal means); brackets mark the differences between stages whose 95% (*), 99% (**) or 99.9% (***) credible interval excludes 0. Points are posterior medians and bars 95% credible intervals.

with the facets: participants who chose environmental impact scored higher on Green (Cohen's $d = 1.18$) and Ethical Science ($d = .80$), and those who chose publication in high impact factor journals lower on Open Science ($d = -.63$).

Reform Across Career Stages

Preregistered hypotheses: training and preregistration decline, sharing rises

Career stage separated knowledge from training and adoption, only partly as hypothesised. Contrary to hypothesis (i), familiarity with open science did not differ

between career stages (permanent staff vs. PhD students: 0.02 [-0.03, 0.06] on the 0–1 scale), and its perceived importance was only slightly higher among PhD students (-0.05 [-0.09, -0.01]). Training, however, was graded: 64% of PhD students, 56% of non-permanent and 44% of permanent researchers had attended a training in open science (permanent staff vs. PhD students: -20 [-28, -11] percentage points). Familiarity with slow science went the other way, being higher among permanent staff (0.10 [0.04, 0.16]).

Adoption depended on the practice (hypothesis ii; Figure 3 A). Sharing outputs was far more common be-

yond the PhD: open access had been used by 46% of PhD students against 81% of non-permanent and 84% of permanent researchers (non-permanent vs. PhD students: 35 [26, 44] percentage points), with similar gaps for open data (33 [24, 43]) and open materials (23 [13, 32]), and for open peer review (27 [18, 36]), which also presupposes taking part in publication. This gap was not only a matter of having something to share: among the participants with at least one publication, PhD students still used open access less often than permanent staff (66% vs. 84%; difference: 18 [8, 28] points). Preregistration followed the opposite gradient: it had been used by 46% of PhD students and 45% of non-permanent researchers, but by 31% of permanent staff (permanent staff vs. PhD students: -15 [-24, -6] points), a third of whom did not know it (35%, vs. 22% of PhD students; difference: 13 [5, 21] points). As permanent staff more often worked in France and in the physical sciences, where preregistration was rare, we examined this gradient within disciplines. It held among social scientists, where preregistration was most common (PhD students: 64%; permanent staff: 50%; difference: -14 [-27, -1] points), including those working in France (-18 [-33, -2]), but not in the life sciences (7 [-7, 21]), and was too imprecise to judge in the physical sciences. Early-career researchers thus do not adopt less open science than their seniors, but a different one: PhD students share fewer outputs, and preregistration is more common before than after obtaining a permanent position.

The levers of adoption (hypotheses iii and iv; Figure 3 B) were structural at every career stage: career recognition was selected by 48–52% of each group, and incentives, funding and time by 37–45%. Contrary to hypothesis (iv), non-permanent researchers did not select time (42%, vs. 43% of permanent staff; difference: 1 [-9, 10] points), a reduced workload (-4 [-12, 4]) or career recognition (4 [-5, 13]) more often than permanent staff. What varied with career stage was the demand for support. The support of senior researchers, the lever most often selected by PhD students, was selected by 54% of them, 39% of non-permanent and 19% of permanent researchers (permanent staff vs. PhD students: -35 [-43, -27] points), and training by 39% of PhD students vs. 21% of permanent staff (-18 [-26, -10]). The pressure of time and evaluation is thus felt at every career stage, and what early-career researchers specifically lack is the support of their seniors.

Facets: procedural reform before a permanent position, ethical concern at entry

The facets mirrored these differences in adoption (Figure 3 C). Non-permanent researchers scored highest on both open science facets: on Open Science, higher than PhD students (0.36 [0.19, 0.53]) and than permanent staff (permanent staff vs. non-permanent: -0.17 [-0.32, -0.01]), and on Rigorous Science, higher than permanent staff (-0.32 [-0.49, -0.15]). Ethical Science was highest among PhD students (non-permanent vs. PhD students: -0.34 [-0.50, -0.18]; permanent staff vs. PhD students: -0.40 [-0.54, -0.26]). Green Science did not differ between career

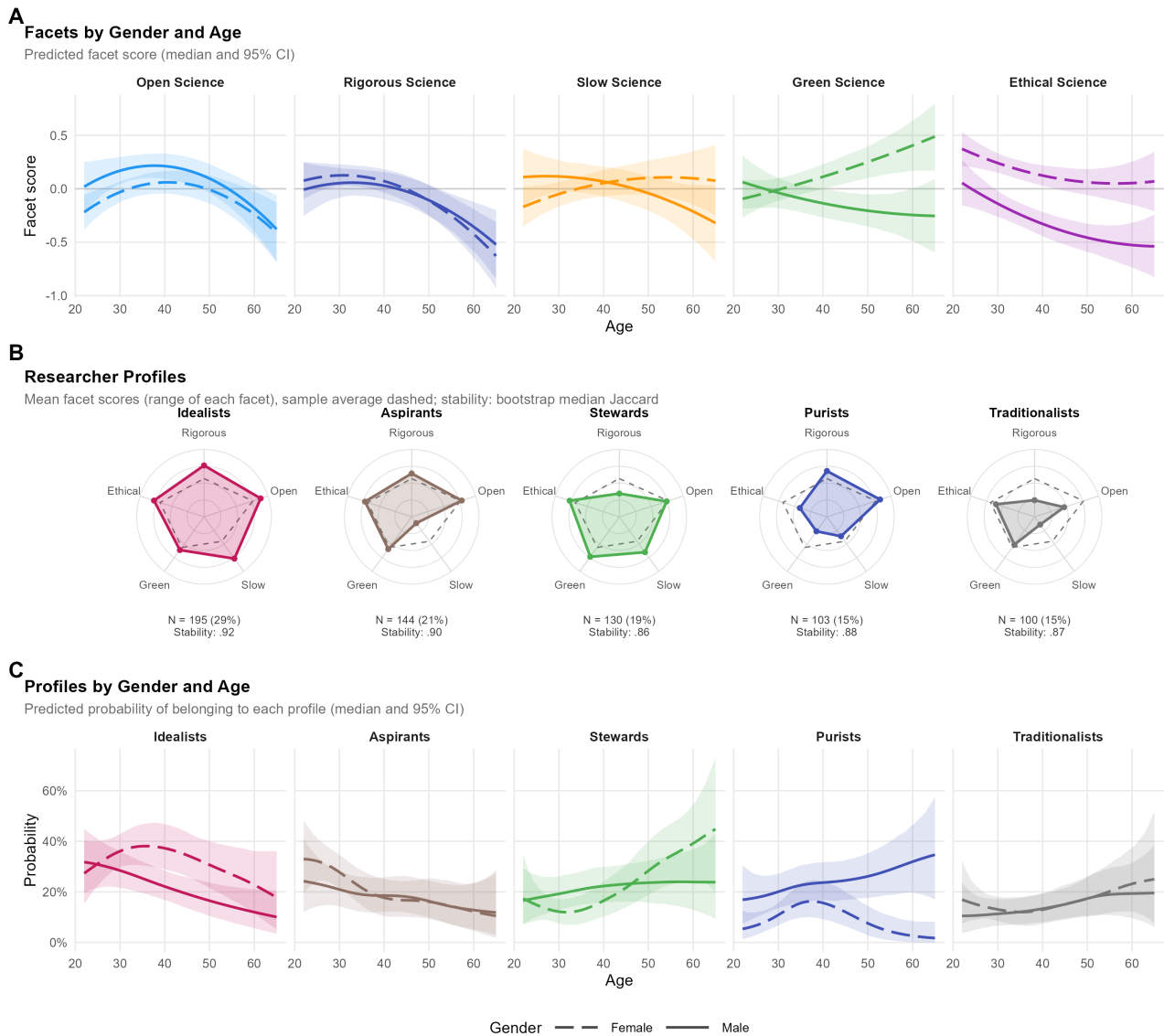
stages, and Slow Science was only slightly higher among permanent staff than among PhD students (0.16 [-0.02, 0.33]); among the participants familiar with slow science, neither the facet (0.00 [-0.17, 0.16]) nor the importance attributed to the movement (-0.02 [-0.08, 0.05] on the 0–1 scale) differed between them. Models adjusting for career worry or for the time available for research gave similar estimates, and neither variable was consistently related to the facets within career stages (supplementary materials). The reform of how research is conducted and shared is thus endorsed most by researchers on fixed-term contracts, whereas ethical concerns are held most strongly at the entry point of research.

Training, knowledge and precariousness

Training and knowledge accounted for part of these differences. Having attended a training in open science (53% of participants) went with higher scores on Open (0.60 [0.47, 0.73]), Rigorous (0.71 [0.58, 0.83]) and Slow Science (0.34 [0.19, 0.48]), adjusting for career stage, and the rarer training in slow science (5%) with higher scores on Slow (1.00 [0.68, 1.31]), Green (0.49 [0.17, 0.80]) and Ethical Science (0.38 [0.12, 0.66]). As PhD students were the most often trained, adjusting for training shrank the Rigorous Science gap between permanent staff and PhD students (from -0.22 [-0.38, -0.06] to -0.09 [-0.23, 0.07]) more than that between permanent and non-permanent researchers (from -0.32 [-0.49, -0.15] to -0.24 [-0.39, -0.08]), and left the stage differences on Open and Ethical Science unchanged. Similarly, among the participants who knew all eight practices, permanent staff no longer differed from PhD students on Rigorous Science (0.01 [-0.12, 0.15]) but still scored lower than non-permanent researchers (-0.17 [-0.30, -0.03]), whereas the higher Ethical Science of PhD students was unchanged (permanent staff vs. PhD students: -0.47 [-0.69, -0.25]; linear models). The lower Rigorous Science of permanent staff thus partly reflects that they were less often trained in, and less often knew, the procedural practices, which is consistent with a cohort effect, although attending a training is also a matter of interest.

Precariousness proper showed no clear relationship with the facets: among the 248 non-permanent researchers, the perceived probability of obtaining a permanent position (0.41 on average) was not clearly related to any facet (all 95% CIs included 0). Nor did endorsing the reforms come at a visible cost to well-being. Of the seven ratings of career perceptions and well-being, adjusting for career stage and for the other facets, only two were clearly related to a facet: Rigorous Science went with feeling more aligned with one's principles (4.1 [2.0, 6.2] points on the 0–100 scale per *SD* of facet score), and Open Science with a higher satisfaction with the quality of one's publications (3.0 [0.6, 5.2]). No facet went with more career worry or with less time for research.

Figure 4
Facets by Gender and Age, and Researcher Profiles



Note. **A:** Facet scores predicted by gender and age (Bayesian linear regressions, second-degree polynomial of age), for women (dashed) and men (solid). **B:** Mean facet scores of the five researcher profiles (k-means), each axis spanning the range of the facet scores; the dashed polygon is the sample average. The Stewards and the Purists take the colour of the facet on which they score highest (Green and Rigorous Science); the three other profiles, which no single facet characterises, take colours outside the facets' palette. Stability is the median Jaccard similarity between the profile and its most similar cluster over 100 bootstrap resamples. **C:** Probability of belonging to each profile predicted by gender and age (Bayesian categorical regression), for women (dashed) and men (solid). Ribbons and bars are 95% credible intervals.

Gender, Discipline and Country

Gender divides the societal facets, increasingly with age

At the average age, men scored lower than women on Ethical Science (-0.44 [-0.57, -0.31]) and Green Science (-0.20 [-0.36, -0.05]), slightly higher on Open Science (0.16 [0.02, 0.30]), and similarly on Rigorous Science (-0.05 [-0.20, 0.10]) and Slow Science (0.05 [-0.12, 0.22]) (Figure 4 A). The gaps on the societal side widened with age. Green Science increased with age among women (0.13 [0.04, 0.21] per decade) but not among men (-0.09

[-0.20, 0.02]), so that the genders did not differ at 25 (0.09 [-0.14, 0.31]) but did at 55 (-0.55 [-0.81, -0.30]), and Ethical Science declined more steeply among men than among women (-0.17 [-0.27, -0.08] vs. -0.10 [-0.17, -0.03] per decade), the gap growing from -0.34 [-0.54, -0.15] at 25 to -0.55 [-0.76, -0.33] at 55. The open science facets, by contrast, followed the same course in both genders, Open Science peaking in mid-career and Rigorous Science declining from mid-career onwards, in line with the differences between career stages. As the data are cross-

sectional, these age trends may reflect differences between generations as much as changes over careers. The gender gaps on the societal facets held among the participants who knew all eight practices (Ethical Science: -0.52 [-0.70, -0.34]; Green Science: -0.21 [-0.41, -0.01]; linear models).

Discipline shapes the epistemic facets

Researchers in the social sciences and humanities scored far higher on Rigorous Science than those in the life sciences (0.52 [0.38, 0.66]) and in the physical sciences and engineering (1.09 [0.92, 1.26]), in line with their use of preregistration (57%, 28% and 8% respectively), and, compared with the physical sciences, higher on Open Science (0.41 [0.23, 0.59]) but slightly lower on Green Science (-0.23 [-0.43, -0.05]). As women were concentrated in the social sciences (63% of these participants, against 34% in the physical sciences), we refitted the gender and age models with discipline as a covariate: the gender gaps on Ethical Science (men vs. women: -0.43 [-0.57, -0.29]) and Green Science (-0.25 [-0.41, -0.09]) were unchanged.

France and other European countries

The 489 respondents working in France differed from the 182 working in other European countries. They had less often attended a training in open science (47% vs. 69%) and far less often preregistered a study (28% vs. 68%), whereas open access was as widespread (71% vs. 73%). Accordingly, they scored lower on Open Science (other European countries vs. France: 0.44 [0.30, 0.58]) and Rigorous Science (0.78 [0.64, 0.92]), higher on Green Science (-0.33 [-0.49, -0.18]), and similarly on Slow and Ethical Science. The respondents from other European countries were mostly non-permanent (23% in permanent positions, vs. 52%) and more often in the social sciences (63% vs. 44%), but the differences held when adjusting for career stage and discipline (Open Science: 0.43 [0.28, 0.58]; Rigorous Science: 0.62 [0.48, 0.76]; Green Science: -0.29 [-0.44, -0.12]). As they were reached through the authors' networks, this comparison describes these respondents rather than their countries.

Five Researcher Profiles

Four combinations of the two sides, and a centre

The methods for choosing the number of clusters favoured two or three clusters (6 methods each), but did so as often in data simulated without groups (on average 5.7 and 8.6 methods over the ten datasets), whereas five clusters were supported by more methods in the observed data (4 of 24) than in any simulated dataset (at most 2). The facet scores were nevertheless no more clustered than the simulated data (average silhouette width with five clusters: .24, vs. .24): the profiles summarise a continuum rather than separate groups. We retained five profiles, the smallest number of clusters at which each combination of the two nearly independent sides formed a profile of its own (with four clusters, the Stewards described below were split between two clusters), and named them after their pattern of facet scores (Figure 4 B). The *Idealists*

(29% of participants) scored above average on all five facets, most of all on Slow Science ($M = 0.91$) and Rigorous Science ($M = 0.71$), and least on Green Science ($M = 0.20$): they endorse the reform of research on every front. The *Aspirants* (21%) were close to average except on Slow Science ($M = -0.95$). The *Stewards* (19%) combined the societal facets (Green Science: $M = 0.73$; Ethical Science: $M = 0.40$) and Slow Science ($M = 0.57$) with a low score on Rigorous Science ($M = -0.80$), Open Science being average ($M = -0.11$), whereas the *Purists* (15%) were nearly their mirror image, above average on both open science facets (Rigorous Science: $M = 0.41$; Open Science: $M = 0.31$) and well below on Green ($M = -1.27$) and Ethical Science ($M = -1.20$). Finally, the *Traditionalists* (15%) scored below average on all facets, most of all on Open ($M = -1.40$) and Rigorous Science ($M = -1.16$), and least on Green Science ($M = -0.22$).

These profiles were reproducible. Over bootstrap resamples, the partition was recovered with a median ARI of .86 [.56, .96], above the median of the data simulated without groups (.70, range: .56–.92), and each profile with a median Jaccard similarity of .86–.92. The partition changed little when the importance ratings of the participants unfamiliar with slow science were treated as missing (ARI = .74). Gaussian mixture models did not offer a better-supported alternative: the best by BIC had four components, one of which was defined by unfamiliarity with slow science, and agreed little with the five profiles (ARI = .44). Coarser partitions are described in the supplementary materials.

The profiles track training, discipline and country

Beyond the facets that define them, the profiles differed in characteristics that were not used in the clustering (Table 1). The *Idealists* were the most often trained in open science (75%, vs. 23–59% in the other profiles) and had most often preregistered a study (69% vs. 2–50%), and most worked in the social sciences (70%). The *Aspirants* were the youngest profile (median age 31 years; 44% PhD students), most often planned to preregister (35% vs. 7–25%) and most often asked for the support of senior researchers (49% vs. 26–35%); 72% had never heard of slow science, at every career stage (68–76% from PhD students to permanent staff). The *Stewards* were mostly permanent staff (62%) working in France (93%). They reported the most changes to their research practices for environmental reasons (55 vs. 13–37 on 0–100) and least often judged limiting the number of publications not feasible (15% vs. 22–44%), but 73% did not know preregistration or registered reports. The *Purists* were the only profile with a majority of men (62%), the least willing to change their practices for environmental reasons (39 vs. 67–86), and the most often opposed to limiting the number of publications (44%). The *Traditionalists* were the least trained in open science (23%), mostly worked in France (93%), and most often chose publication in high impact factor journals as a criterion of quality (17% vs. 4–9%). These characteristics are described one at

Table 1
Characteristics of the Five Researcher Profiles

Characteristic	Idealists	Aspirants	Stewards	Purists	Traditionalists
<i>N</i>	195	144	130	103	100
Median age (years)	33	31	40	38	40
Women (%)	66	62	54	37	59
PhD students (%)	31	44	21	19	27
Permanent position (%)	37	29	62	49	51
No publication (%)	15	22	10	11	18
Social sciences and humanities (%)	70	54	28	55	23
Working in France (%)	64	62	93	59	93
Trained in open science (%)	75	54	37	59	23
Trained in slow science (%)	12	1	5	4	0
Not at all familiar with slow science (%)	1	72	8	39	64
Preregistration used (%)	69	47	2	50	5
Preregistration planned (%)	25	35	15	21	7
Registered reports planned (%)	55	53	11	39	7
Preregistration or registered reports unknown (%)	7	23	73	20	84
Lever: support from senior researchers (%)	35	49	26	28	33
Lever: training (%)	24	31	27	21	37
Criterion: high impact factor journals (%)	5	6	4	9	17
Criterion: originality (%)	30	42	46	48	51
Criterion: significance (%)	35	32	38	51	53
Criterion: environmental impact (%)	6	6	16	2	3
Limiting publications judged not feasible (%)	22	26	15	44	28
Changed research practices for environmental reasons	36	37	55	13	35
Willing to change practices for environmental reasons	79	80	86	39	67
Importance of team diversity	79	77	76	45	62

Note. Characteristics not used to define the profiles. Percentages are of the participants of each profile; the last three rows are mean ratings on the 0–100 scale. Criteria are those selected among the three most important criteria of scientific quality; levers are those selected among the (up to five) avenues that would help adopting open science practices.

a time, without adjustment for one another; discipline and country, in particular, predicted profile membership strongly ($\chi^2(12) = 164.0, p < .001$, for adding both to a multinomial model with gender and age).

Men are more often Purists, at every age

Profile membership also varied with gender and age (Figure 4 C). Men were more likely than women to be Purists (at 25: 11 [2, 21] percentage points; at 55: 24 [13, 36]), and between 25 and 55, women became more likely to be Stewards (19 [6, 33]) and less likely to be Aspirants (-18 [-30, -5]), whereas men became less likely to be Idealists (-17 [-30, -3]). In multinomial models adjusted for discipline and country, the gender gap on the Purists (at 55: 30 [19, 41]) and the decline of the Aspirants among women (-13 [-26, -1]) remained, whereas the rise of the Stewards among women (11 [-1, 22]) and the decline of the Idealists among men (-13 [-28, 1]) became uncertain.

Discussion

This study is, to our knowledge, the first to survey open, slow, green and ethical science together, among researchers at all career stages working in Europe, mostly in France. Three findings stand out. First, the four move-

ments do not correspond to four dimensions of researchers' attitudes and practices, but to five facets organised in two nearly independent sides: an epistemic side, concerned with how research is conducted and shared (Open and Rigorous Science), and a societal side, concerned with its consequences (Green and Ethical Science), loosely bridged by Slow Science. Second, career stage goes with *which* reforms researchers carry more than with how much they endorse reform, and the barriers to open science are structural at every stage. Third, gender divides the societal side, increasingly with age, and five researcher profiles summarise the ways in which the two sides are combined.

Of the four preregistered hypotheses, the first was supported for training and for the knowledge of specific practices, but not for familiarity with the movement: early-career researchers had more often been trained in open science and were less often unfamiliar with preregistration, but knew the movement no better than their seniors. The second predicted that this advantage would not extend to adoption; adoption, rather, differed by practice in opposite directions, PhD students sharing fewer outputs and preregistration being more common before than after obtaining a permanent position. The third was supported for institutional support but only partly for time, in the form

of levers rather than barriers: institutional levers (career recognition, and incentives from funders and institutions) were the most selected, whereas time came fourth, close behind dedicated funding. The fourth was not: the levers related to time and evaluation were selected as often by permanent as by non-permanent researchers.

Two sides of reform-oriented science

The movements did not map onto the facets. Open science split into the sharing of outputs (Open Science) and the reform of procedures (Rigorous Science), and the green and ethical movements partly merged, the beliefs about the environmental responsibility of research going with the care for its societal consequences and for the diversity of its teams. Movements are frameworks of advocacy, which bundle practices under a common label. Researchers organise their commitments differently, according to what the practices demand of them and to the diagnosis they share.

We expected the attitudes towards the four movements to be positively correlated. They were, but only within each side: across sides, the facets were nearly independent. The two sides correspond to two diagnoses of what is wrong with research: its reliability, which is the concern of the replication crisis, and its responsibility towards society, the planet and the people who do research. The same divide runs through the literature on reform. Within open science itself, some schools of thought seek to make research more efficient and measurable, others to make knowledge accessible and democratic (Fecher & Friesike, 2014), and responsible research and innovation defines responsibility by the purposes and consequences of research rather than by its reliability (Owen et al., 2012). Our results suggest that researchers, too, respond to one diagnosis or the other more than to both.

The two sides were independent rather than opposed. The conflicts anticipated in the Introduction (between preprints and publication volume, article processing charges and inclusion, or international collaboration and carbon footprint) could not be tested, as no item measured these trade-offs, and independent attitudes are compatible with conflicting decisions. More telling is that for about a third of the participants, the Purists and the Stewards, the two sides are held apart: one can reform one's procedures and set the societal side aside, or embrace the responsibility of research without reforming its procedures.

The criteria of scientific quality tell the same story. Rigour was the consensus criterion and journal prestige was not, which suggests that the critique of journal metrics has reached the researchers themselves (San Francisco Declaration on Research Assessment, 2012). Environmental impact and inclusivity, although rated as important elsewhere in the survey, were rarely chosen as criteria of *quality*: the societal concerns are seen as the ethics of research rather than as part of its quality.

These results have an implication for the reform of research assessment. The current reforms (Coalition for Advancing Research Assessment, 2022; San Francisco

Declaration on Research Assessment, 2012) focus on how research is evaluated, through the quality of its process rather than the prestige of its outlets. They may carry the epistemic side of reform-oriented science, but there is no reason to expect them to carry the societal side with it. Integrated frameworks, such as responsible research and innovation (Owen et al., 2012), need to make the link between the two sides explicit rather than assume it.

Slow science: a bridge built partly on knowledge

Slow science was the least known of the four movements, so that the Slow Science facet measures engagement with it, knowing it as much as valuing it. Among the participants who knew it, its link with the epistemic side shrank or vanished, whereas its link with the societal side held. As researchers understand it, slow science is thus closer to the societal side than its origin in the debates on research quality would suggest (Frith, 2020; Stengers, 2018), and closer to the critiques of the accelerated academy, which tie the pace of research to the conditions of those who do it (Berg & Seeber, 2016; Mountz et al., 2015).

Researchers understand slow science as quality over quantity, and as a change in how research is valued, to be carried by institutions rather than by individuals: every measure was judged more feasible at the institutional than at the individual level, and the measures that cap output, such as limiting the number of publications per year (Frith, 2020), were the most contested. Policies inspired by slow science may thus find more support if they act on assessment metrics and timescales than if they ask individuals to publish less.

Slow academia has been criticised as a privilege of tenured academics, whose careers no longer depend on their output (Berg & Seeber, 2016), a criticism that its proponents have answered by framing slowing down as a collective rather than an individual act (Mountz et al., 2015). Our data do not support the privilege reading. Slow Science was only slightly higher among permanent staff, it was unrelated to perceived precariousness and career worry, and among the participants familiar with the movement, career stage made no difference to it. What sets researchers apart on slow science is exposure rather than security, as the Aspirants illustrate: they differ from the Idealists mainly by not knowing slow science, at every career stage. Making the movement known may thus do more for its support than securing positions would.

Open science across career stages

Sharing research outputs is now mainstream among the researchers surveyed, whereas the reforms of procedures (preregistration, registered reports, replication and open peer review) are much rarer, a gap that reflects unfamiliarity and unrealised intentions more than rejection. The two kinds of practices differ in their institutional backing. Open access and data sharing are supported by mandates from funders and national policies (Guedj & Ramjoué, 2015; Ministère de l'Enseignement supérieur,

de la Recherche et de l'Innovation, 2021); the reforms of procedures are not, and remain bound to disciplines. Which practices spread may thus depend on which are mandated (Norris et al., 2022). The comparison between France and other European countries points the same way: open access was as widespread in France, where national policy centres on publications and their deposit in the open archive HAL (Ministère de l'Enseignement supérieur, de la Recherche et de l'Innovation, 2021), whereas preregistration was far rarer. Participants, for their part, asked for structural levers (career recognition, incentives, funding and time) more than for information and training, as in previous surveys (Pownall et al., 2023; Toribio-Flórez et al., 2021; Zečević et al., 2020). Yet a sizeable minority did not know preregistration or registered reports, more often among permanent staff: information about the procedural practices is still needed, even if it is not asked for.

Early-career researchers know the movement no better than their seniors, but have more often been trained in it and know its procedures better, possibly because training has only recently entered doctoral programmes. Nor do they adopt less open science: they adopt a different one. PhD students share fewer outputs, even among those with a publication, which suggests that the decision to share rests with their seniors (Allen & Mehler, 2019). Preregistration, conversely, was more common before than after obtaining a permanent position, also within the social sciences and among those working in France, which is consistent with a cohort effect: the researchers who entered research after preregistration became common in their field preregister more. These differences coincide with differences in discipline and country (see the Limitations), but they nuance the portrayal of early-career researchers as the drivers of open science (Farnham et al., 2017; Nicholas et al., 2019): they are ahead in the reform of procedures rather than in the sharing of outputs. Ethical Science, finally, was highest among PhD students, a generational shift or an effect of socialisation into academia that cross-sectional data cannot tell apart.

The levers related to time and career recognition were selected as often by permanent as by non-permanent researchers, and neither perceived precariousness nor career worry was related to the facets: the pressure of time and evaluation is system-wide rather than specific to precarious positions. Nor did endorsing the reforms show any sign of a cost to well-being, although cross-sectional data cannot show that it improves it. What early-career researchers specifically lack is the support of their seniors, which half of the PhD students asked for. Yet permanent staff were the least trained in open science and the least engaged in the reform of procedures, so that this support may have to come from those least equipped to give it. As supervisors who practise open science are more likely to have PhD candidates who do (Haven et al., 2023), training and recognising supervisors could be a direct lever for the adoption of procedural reforms, a suggestion that follows from these requests rather than from a test.

Gender and the societal side of reform

Women scored higher than men on Ethical and Green Science, whereas the genders differed little on the open science facets: the gender divide lies on the societal side of reform-oriented science, not on the epistemic one. The gaps widened with age, and were explained neither by discipline, although women were concentrated in the social sciences, nor by the knowledge of the practices, consistent with the gender differences reported in environmental concern (e.g., Zelezny et al., 2000). Among the profiles, men were more often Purists, who pursue the reform of procedures without the societal side, at every age and with or without adjustment for discipline and country, whereas the greater share of Stewards among older women partly reflected the disciplines and countries in which they worked. As with career stage, ageing and cohort cannot be told apart in these data, and the comparison between genders rests on a measurement model whose invariance across genders was only borderline (see the Limitations). One implication is that defining good research by its reliability alone, as the procedural reforms tend to do, would leave aside commitments that are more often held by women.

Five researcher profiles

The profiles summarise a continuum rather than separate groups: the facet scores were no more clustered than data simulated without groups, and mixture models preferred a solution organised around familiarity with slow science. Five is a choice of resolution, the smallest at which every combination of the two nearly independent sides forms a profile of its own: both sides (the Idealists), the epistemic side only (the Purists), the societal side only (the Stewards) and neither (the Traditionalists), around a centre that differs mainly by not knowing slow science (the Aspirants). The profiles are thus a vocabulary for combinations of facets, not types of researchers. Their names describe attitudes, not age or gender, and they partly reflect the national and disciplinary make-up of the sample, the Stewards and the Traditionalists mostly working in France, and the Idealists mostly in the social sciences.

Their other characteristics (Table 1), described one at a time, suggest hypotheses rather than conclusions. The Idealists, the most often trained and mostly social scientists, may owe their integrated view of reform to training and to the disciplines at the centre of the replication crisis; they are natural allies for integrated reform agendas. The Aspirants, the youngest profile, reported more intentions than practices and asked most for the support of senior researchers, so that whether their plans become practice may depend on their supervisors. The Stewards were the most accepting of caps on output, in line with a sufficiency conception of academic work (Kostakis, 2025), but most did not know preregistration or registered reports, so that their low Rigorous Science may reflect their disciplines and national context more than a rejection of procedural reform; for them, the reform of procedures could be framed as part of a responsible science, transparency being owed to participants and to society. The Purists may

hold a value-free ideal of science (e.g., Douglas, 2009), in which reform serves the reliability of research and remains compatible with an evaluation based on output and impact; for them, sustainability and inclusion could be framed as part of rigour, through the diversity of samples and the generalisability of findings. The Traditionalists combined unfamiliarity with the procedural reforms and an attachment to journal prestige as a criterion of quality, so that training alone may not be enough to reach them.

To our knowledge, no previous study has grouped researchers by their combination of reform-oriented attitudes, even within open science: surveys have described how widely each practice is used and endorsed (Christensen et al., 2019; Ferguson et al., 2023), not how researchers combine them. The profiles offer a first vocabulary, to be tested in other samples and with longitudinal data.

Limitations

Several aspects of the study departed from the preregistration. The sample fell short of the preregistered target of 3,000 participants. The barriers to open science, the object of hypotheses (iii) and (iv), were measured as the levers that would help participants adopt it. The analyses of the facets, of the profiles and of well-being were exploratory, and the preregistered comparison with published estimates from other countries remains to be done. Recruitment through networks and mailing lists, and through researchers working on these movements, likely over-represented researchers interested in reform, as the importance given to open science suggests ($M = 82.1$, $SD = 22.6$ on 0–100). Most participants worked in France. The percentages reported here describe the respondents, not the population of European researchers, and the adoption of the practices is self-reported, with the social desirability and recall biases this entails. The sample included too few non-binary participants ($n = 9$) to analyse them separately.

The structure of the facets rests on few items and on a single sample. Ethical Science rests on two items, plus the two environmental beliefs that also load on it, and Slow Science on two, and the cross-loadings of the retained CFA were suggested by the EFA of the same data, so that this structure needs replication in an independent sample. It also partly coincides with the format of the items, the epistemic side resting mainly on the knowledge and use of practices and the societal side on 0–100 ratings. However, the importance of open science, the only such rating on the epistemic side, correlated with the green and ethical ratings ($r = .01$ to $.18$) about as much as with the practices ($r = .08$ to $.29$), whereas the practices and the green and ethical ratings were nearly unrelated (mean absolute $r = .05$), which argues against a mere artefact of format.

Two facets mix knowledge and endorsement. Slow Science rests on the familiarity with the movement and its importance, two strongly correlated items ($r = .79$), the second of which nearly all unfamiliar participants rated 0. Rigorous Science, as unfamiliarity was scored below

deliberate non-use, partly reflects not knowing preregistration and registered reports: its scores correlated only .59 with those obtained when unfamiliarity was treated as missing, against .99 among the participants who knew all eight practices. Group differences on these facets, and the profiles that rest on them, therefore combine knowing and endorsing the reforms. The analyses restricted to familiar participants suggest that this matters for the links of Slow Science with the epistemic side and for the Rigorous Science gap between permanent staff and PhD students, but not for the societal facets or for the gender gaps.

Career stage and age are largely confounded, as in any cross-sectional survey of researchers, so that the models on age and on career stage describe the same gradient from two angles, and the data cannot tell apart what is due to age or generation from what is due to position and employment. Career stage was also confounded with discipline and country. Permanent staff more often worked in France (86%, vs. 62% of PhD students and 64% of non-permanent researchers) and in the physical sciences and engineering (30%, vs. 10% and 9%), both of which were strongly related to Rigorous Science and to preregistration, and the sample was too small to separate career stage from them with useful precision. The preregistration gradient held within the social sciences, including in France, but the Rigorous Science gap between permanent staff and PhD students was not clear among social scientists (-0.09 [$-0.29, 0.11$]), and may partly reflect the disciplinary and national make-up of each stage. The profiles, similarly, partly reflect discipline and country.

The comparisons of facet scores between groups assume that the items measure the facets in the same way in each group. Metric invariance (equal loadings) held across career stages ($\Delta CFI = -.004$) but was borderline across genders ($-.016$) and disciplines ($-.013$), and scalar invariance (equal intercepts) held for none of the groupings (ΔCFI from $-.016$ to $-.049$). The mean differences between genders and between disciplines should therefore be read as differences in facet scores computed from a common model, and part of them may reflect differences in how the items were understood rather than in the facets themselves.

Finally, the respondents working outside France were reached through the authors' networks, so that the comparison between France and other European countries describes these respondents rather than their countries; discipline was measured in three broad domains only; and most analyses beyond the four hypotheses were exploratory and not corrected for multiple comparisons, so that patterns that recur across models deserve more weight than any single interval.

Conclusion

Reform-oriented science is plural. Researchers commit to the reliability of research and to its responsibility separately, and the boundaries of the movements do not match the way researchers organise their values: open science splits into the sharing of outputs and the reform

of procedures, and the green and ethical concerns merge into a sense of the responsibility of research. The barriers to open science are structural at every career stage, and early-career researchers ask for the support of their seniors, so that reforming research assessment, and possibly training supervisors, appear as the main levers. Integrated reform agendas, finally, need to build the bridge between reliability and responsibility rather than assume it; the Idealists, 29% of the participants, show that the two can be held together.

References

- Allen, C., & Mehler, D. M. A. (2019). Open science challenges, benefits and tips in early career and beyond. *PLoS Biology*, 17(5), e3000246. <https://doi.org/10.1371/journal.pbio.3000246>
- Berg, M., & Seeber, B. K. (2016). *The slow professor: Challenging the culture of speed in the academy*. University of Toronto Press. <https://doi.org/10.3138/9781442663091>
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1), 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Cabello Valdes, C., Rentier, B., Kaunismaa, E., Metcalfe, J., Esposito, F., McAllister, D., Maas, K., Van de Velde, K., & O’Carroll, C. (2017). *Evaluation of research careers fully acknowledging open science practices: Rewards, incentives and/or recognition for researchers practicing open science*. European Commission, Directorate-General for Research; Innovation. <https://doi.org/10.2777/75255>
- Charrad, M., Ghazzali, N., Boiteau, V., & Niknafs, A. (2014). NbClust: An R package for determining the relevant number of clusters in a data set. *Journal of Statistical Software*, 61(6), 1–36. <https://doi.org/10.18637/jss.v061.i06>
- Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal*, 14(3), 464–504. <https://doi.org/10.1080/10705510701301834>
- Christensen, G., Wang, Z., Levy Paluck, E., Swanson, N., Birke, D. J., Miguel, E., & Littman, R. (2019). *Open science practices are on the rise: The State of Social Science (3S) survey*. MetaArXiv. <https://doi.org/10.31222/osf.io/5rksu>
- Coalition for Advancing Research Assessment. (2022). *Agreement on reforming research assessment*. <https://coara.eu/agreement/the-agreement-full-text/>
- Douglas, H. E. (2009). *Science, policy, and the value-free ideal*. University of Pittsburgh Press.
- Enright, B., & Facer, K. (2017). Developing reflexive identities through collaborative, interdisciplinary and precarious work: The experience of early career researchers. *Globalisation, Societies and Education*, 15(5), 621–634. <https://doi.org/10.1080/14767724.2016.1199319>
- Farnham, A., Kurz, C., Öztürk, M. A., Solbiati, M., Myllyntaus, O., Meekes, J., Pham, T. M., Paz, C., Langiewicz, M., Andrews, S., Kanninen, L., Agbembiese, C., Guler, A. T., Durieux, J., Jasim, S., Viessmann, O., Frattini, S., Yembergenova, D., Benito, C. M., ... Hettne, K. (2017). Early career researchers want open science. *Genome Biology*, 18, 221. <https://doi.org/10.1186/s13059-017-1351-7>
- Fecher, B., & Friesike, S. (2014). Open science: One term, five schools of thought. In S. Bartling & S. Friesike (Eds.), *Opening science: The evolving guide on how the internet is changing research, collaboration and scholarly publishing* (pp. 17–47). Springer. https://doi.org/10.1007/978-3-319-00026-8_2
- Ferguson, J., Littman, R., Christensen, G., Paluck, E. L., Swanson, N., Wang, Z., Miguel, E., Birke, D., & Pezzuto, J.-H. (2023). Survey of open science practices and attitudes in the social sciences. *Nature Communications*, 14, 5401. <https://doi.org/10.1038/s41467-023-41111-1>
- Fox, J., Pearce, K. E., Massanari, A. L., Riles, J. M., Szulc, Ł., Ranjit, Y. S., Trevisan, F., Soriano, C. R. R., Vitak, J., Arora, P., Ahn, S. J. (Grace), Alper, M., Gambino, A., Gonzalez, C., Lynch, T., Williamson, L. D., & Gonzales, A. L. (2021). Open science, closed doors? Countering marginalization through an agenda for ethical, inclusive research in communication. *Journal of Communication*, 71(5), 764–784. <https://doi.org/10.1093/joc/jqab029>
- Frith, U. (2020). Fast lane to slow science. *Trends in Cognitive Sciences*, 24(1), 1–2. <https://doi.org/10.1016/j.tics.2019.10.007>
- Guedj, D., & Ramjoué, C. (2015). European Commission policy on open-access to scientific publications and research data in Horizon 2020. *Biomedical Data Journal*, 1(1), 11–14. <https://doi.org/10.11610/bmdj.01102>
- Hartigan, J. A., & Wong, M. A. (1979). Algorithm AS 136: A K-means clustering algorithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 28(1), 100–108. <https://doi.org/10.2307/2346830>
- Haven, T. L., Abunijela, S., & Hildebrand, N. (2023). Biomedical supervisors’ role modeling of open science practices. *eLife*, 12, e83484. <https://doi.org/10.7554/eLife.83484>
- Hennig, C. (2007). Cluster-wise assessment of cluster stability. *Computational Statistics & Data Analysis*, 52(1), 258–271. <https://doi.org/10.1016/j.csda.2006.11.025>
- Herschberg, C., Benschop, Y., & Brink, M. van den. (2018). Precarious postdocs: A comparative study on recruitment and selection of early-career researchers. *Scandinavian Journal of Management*, 34(4), 303–310. <https://doi.org/10.1016/j.scaman.2018.10.001>
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218. <https://doi.org/10.1007/BF01908075>
- Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLoS Medicine*, 2(8), e124. <https://doi.org/10.1371/journal.pmed.0020124>
- Ioannidis, J. P. A., Fanelli, D., Dunne, D. D., & Goodman, S. N. (2015). Meta-research: Evaluation and improve-

- ment of research methods and practices. *PLoS Biology*, 13(10), e1002264. <https://doi.org/10.1371/journal.pbio.1002264>
- John, L. K., Loewenstein, G., & Prelec, D. (2012). Measuring the prevalence of questionable research practices with incentives for truth telling. *Psychological Science*, 23(5), 524–532. <https://doi.org/10.1177/0956797611430953>
- Kostakis, V. (2025). Are we practising what we preach? The degrowth academic trap. *Degrowth Journal*, 3. <https://doi.org/10.36399/Degrowth.003.01.12>
- Lilienfeld, S. O. (2017). Psychology's replication crisis and the grant culture: Righting the ship. *Perspectives on Psychological Science*, 12(4), 660–664. <https://doi.org/10.1177/1745691616687745>
- Lüdecke, D., Ben-Shachar, M. S., Patil, I., & Makowski, D. (2020). Extracting, computing and exploring the parameters of statistical models using R. *Journal of Open Source Software*, 5(53), 2445. <https://doi.org/10.21105/joss.02445>
- Makowski, D., Ben-Shachar, M. S., Wiernik, B. M., Patil, I., Thériault, R., & Lüdecke, D. (2025). modelbased: An R package to make the most out of your statistical models through marginal means, marginal effects, and model predictions. *Journal of Open Source Software*, 10(109), 7969. <https://doi.org/10.21105/joss.07969>
- Mariette, J., Blanchard, O., Berné, O., Aumont, O., Carrey, J., Ligozat, A.-L., Lellouch, E., Roche, P.-E., Guennebaud, G., Thanwerdas, J., Bardou, P., Salin, G., Maigne, E., Servan, S., & Ben-Ari, T. (2022). An open-source tool to assess the carbon footprint of research. *Environmental Research: Infrastructure and Sustainability*, 2(3), 035008. <https://doi.org/10.1088/2634-4505/ac84a4>
- Ministère de l'Enseignement supérieur, de la Recherche et de l'Innovation. (2021). *Deuxième plan national pour la science ouverte: Généraliser la science ouverte en France, 2021–2024*. Ministère de l'Enseignement supérieur, de la Recherche et de l'Innovation. <https://www.ouvri.lascience.fr/deuxieme-plan-national-pour-la-science-ouverte/>
- Mountz, A., Bonds, A., Mansfield, B., Loyd, J., Hyndman, J., Walton-Roberts, M., Basu, R., Whitson, R., Hawkins, R., Hamilton, T., & Curran, W. (2015). For slow scholarship: A feminist politics of resistance through collective action in the neoliberal university. *ACME: An International Journal for Critical Geographies*, 14(4), 1235–1259. <https://doi.org/10.14288/acme.v14i4.1058>
- Nicholas, D., Watkinson, A., Boukacem-Zeghmouri, C., Rodríguez-Bravo, B., Xu, J., Abrizah, A., Świgoń, M., Clark, D., & Herman, E. (2019). So, are early career researchers the harbingers of change? *Learned Publishing*, 32(3), 237–247. <https://doi.org/10.1002/leap.1232>
- Norris, E., Clark, K., Munafò, M. R., Jay, C., Baldwin, J., Lautarescu, A., Pedder, H., Page, M., Rinke, E. M., Burn, C., Cawthorn, W. P., Ballou, N., Glover, S., Evans, S., Rossit, S., Soltanlou, M., Wise, E. L., Kelson, M., Soliman, N., ... Pennington, C. R. (2022). *Development of the Brief Open Research Survey (BORS) to measure awareness and uptake of open research practices*. MetaArXiv. <https://doi.org/10.31222/osf.io/w48yh>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- Owen, R., Macnaghten, P., & Stilgoe, J. (2012). Responsible research and innovation: From science in society to science for society, with society. *Science and Public Policy*, 39(6), 751–760. <https://doi.org/10.1093/scipol/scs093>
- Pontika, N., Knoth, P., Cancellieri, M., & Pearce, S. (2015). Fostering open science to research using a taxonomy and an eLearning portal. *Proceedings of the 15th International Conference on Knowledge Technologies and Data-Driven Business (i-KNOW '15)*, 1–8. <https://doi.org/10.1145/2809563.2809571>
- Pownall, M., Terry, J., Collins, E., Sladekova, M., & Jones, A. (2023). UK psychology PhD researchers' knowledge, perceptions, and experiences of open science. *Cogent Psychology*, 10(1), 2248765. <https://doi.org/10.1080/23311908.2023.2248765>
- Revelle, W. (2026). *Psych: Procedures for psychological, psychometric, and personality research*. Northwestern University. <https://CRAN.R-project.org/package=psych>
- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- San Francisco Declaration on Research Assessment. (2012). *San Francisco declaration on research assessment*. <https://sfdora.org/read/>
- Stengers, I. (2018). *Another science is possible: A manifesto for slow science*. Polity Press.
- Toribio-Flórez, D., Anneser, L., deOliveira-Lopes, F. N., Pallandt, M., Tunn, I., & Windel, H. (2021). Where do early career researchers stand on open science practices? A survey within the Max Planck Society. *Frontiers in Research Metrics and Analytics*, 5, 586992. <https://doi.org/10.3389/frma.2020.586992>
- Zečević, K., Houghton, C., Noone, C., Lee, H., Matvienko-Sikar, K., & Toomey, E. (2020). Exploring factors that influence the practice of open science by early career health researchers: A mixed methods study. *HRB Open Research*, 3, 56. <https://doi.org/10.12688/hrbopenres.13119.1>
- Zelezny, L. C., Chua, P.-P., & Aldrich, C. (2000). New ways of thinking about environmentalism: Elaborating on gender differences in environmentalism. *Journal of Social Issues*, 56(3), 443–457. <https://doi.org/10.1111/0022-4537.00177>